

Iniziato	Friday, 21 June 2024, 09:23
Stato	Completato
Terminato	Friday, 21 June 2024, 09:30
Tempo impiegato	7 min. 1 secondo

Domanda **1**

Risposta corretta

Punteggio max.: 2,00

Let f, g be the functions defined as $f(n) = n10^4 n^n \log^3 n$ and $g(n) = 10^5 n^{n+1} \log^3 n$.

Scegli una o più alternative:

- ☒ $f \in \Theta(g)$ ✓
- ☒ $f \in O(g)$ ✓
- ☒ $f \in \Omega(g)$ ✓

Le risposte corrette sono: $f \in O(g), f \in \Omega(g), f \in \Theta(g)$

Domanda **2**

Risposta corretta

Punteggio max.: 2,00

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- ☒ Rational numbers ✓
- ☐ Infinite sequences of bits
- ☐ Complex numbers
- ☒ Finite sequences of natural numbers ✓

Le risposte corrette sono: Rational numbers, Finite sequences of natural numbers

Domanda **3**

Risposta errata

Punteggio max.: 2,00

The problem of checking whether a Turing machine rejects the string 00101 is:

Scegli una o più alternative:

- ☒ In the class **NP**
- ☐ Not in the class **P**
- ☐ Undecidable.
- ☒ Decidable in linear time

✗

✗

Le risposte corrette sono: Undecidable., Not in the class **P**

Domanda **4**

Parzialmente corretta

Punteggio max.: 2,00

Suppose a friend of yours come to you saying he found a language $\mathcal{L}$ such that

$\text{SAT} \subseteq \mathcal{L}$ and which is decidable in polynomial time. You can conclude that:

Scegli una o più alternative:

- ☐ Your friend cannot be right.
- ☒ Your friend has proved that $\mathbf{P} = \mathbf{NP}$.
- ☒ Your friend is possibly right.
- ☐ Your friend has proved that $\mathbf{P} \neq \mathbf{NP}$.

✗

✓

La risposta corretta è: Your friend is possibly right.

Domanda **5**

Risposta errata

Punteggio max.: 2,00

The concept classes of axis-aligned squares in $\mathbb{R}^2$ is:

Scegli una o più alternative:

- ☒ PAC learnable, but not efficiently so.
- ☐ Not even PAC learnable
- ☒ PAC learnable
- ☐ Efficiently PAC learnable.

✗

✓

Le risposte corrette sono: PAC learnable, Efficiently PAC learnable.

Vai a...



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